

Stakeholder

14+3 · WEIGHT 2 (Lukas Moroz)

Buy the foundational brick → V-JEPA 2-AC. <72h H/14. Dual semantic+local. Open code. 3-way taxonomy. Decide: required for world-model track.

Spine: Predict embeddings, not pixels · multi-block + EMA + narrow predictor · JEA|MAE|I-JEPA ·
→V-JEPA→2-AC (cite 2-AC for robots) · 54.2 / 66.9>>40.7 / ~10× / 72.4

Decision: required reading for world-model track · optional for VLA backbone bake-offs

Talk arc — Stakeholder

Buy → proof → risks → decision

02-04

Why buy + taxonomy

brick → 2-AC · 3-way SSL · dual competence

05-06

Headlines + compute

79.3 / 73.3 / 81.1 · Fig.1 · <72h

07-09

Risks + market

image-only · FT -0.7pp · VLAs chose SigLIP

10-12

Decision + lineage

required WM · optional VLA · cite 2-AC

13-14

Tattoo + close

exam bite

Why buy — five reasons to schedule

Accompanying Models · world-model track

Foundational brick

Non-generative SSL that Meta later scaled into video world models + robot planning (V-JEPA 2-AC).

Practical compute

ViT-H/14 <72 h on 16 A100; linear 79.3%; 1% 73.3%; no view-aug engineering.

Dual competence

Semantic (closes gap to DINO/iBOT) AND local (Clevr Dist 72.4 >> DINO 53.4).

Open recipe

facebookresearch/ijepa — clear to reimplement / ablate.

Taxonomy clarity

Contrastive / generative / predictive — kills 'SSL = contrastive' confusion.

3-way taxonomy — conceptual product

Contrastive JEA | Generative MAE | Predictive I-JEPA

JEA / contrastive

SimCLR · BYOL · DINO · IBOT

Match embeddings of
augmented views

Hand-crafted augs
multi-view cost

Generative / MIM

MAE · BEiT · SimMIM

Reconstruct missing
pixels / tokens

Strong FT
weaker linear

JEPA (this)

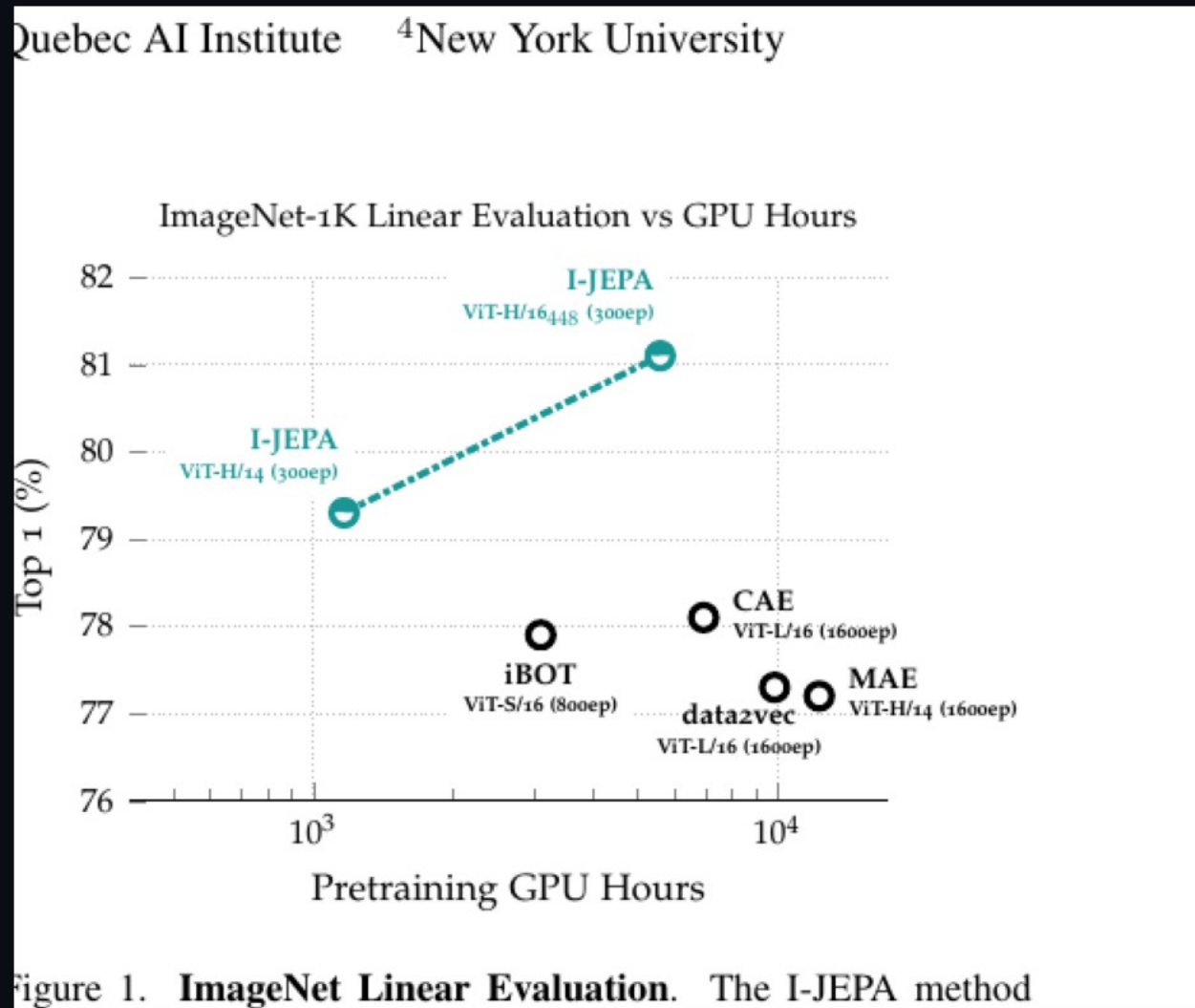
I-JEPA → V-JEPA → 2-AC

Predict target
embeddings

Semantic + local
single view

Headlines + compute — Fig. 1

79.3% H/14 · 81.1% H/16@448 · ~10× cheaper than MAE H/14



Paper Fig. 1 — ImageNet linear vs GPU hours. Cite <1200 GPU-h / 16×A100 <72h from abstract §1 §7.

Dual competence — semantic + local

Less invariance over-bias than DINO/iBOT

87.5

CIFAR100 linear

58.4

Places205

47.6

iNat18

86.7

Clevr Count

72.4

Clevr Dist
= MAE; $\gg$ DINO 53.4

79.3%

IN linear H/14

Risks / costs / limits

Be honest with the buyer

Image-only paper

No planning / action / robot numbers here — value is lineage + idea, not a drop-in world model.

FT still **−0.7 pp**

87.1 vs MAE 87.8 (H@448) despite 5.3× fewer epochs — if KPI is end-to-end FT only, MAE fine.

Mask hypersensitivity

Tables 6–10: wrong multi-block → collapse to mid-teens 1% accuracy.

~7% slower / iter

Savings from epoch count, not magic FLOPs vs MAE.

VLAs chose SigLIP/DINO

π_0 , OpenVLA, GR00T did not adopt I-JEPA encoders — robotics payoff is world-model (2-AC).

Scale to match augs

Need H/16@448 to match iBOT-level linear narratives.

Market reality — VLAs did not pick I-JEPA

Vision-tower market $\neq$ world-model track

π_0 / $\pi_{0.5}$	SigLIP $\rightarrow$ PaliGemma + FM expert	TENTATIVE rhyme only
OpenVLA	SigLIP + DINOv2 fusion	No JEPA cite
GR00T-N1	Eagle / dual-system	No JEPA cite
BEHAVIOR'25 winners	$\pi_{0.5}$ + FM + SigLIP	Parallel bet
JEPA robotics today	V-JEPA 2-AC planning	CLEAR — cite 2506.09985

Decision ask

What to do after this paper club

REQUIRED

Treat I-JEPA as required reading for the world-model track. It is the static-image birth certificate of the line that becomes V-JEPA 2-AC robot planning.

OPTIONAL

Optional for VLA backbone bake-offs unless you are evaluating latent predictors. Default VLA trunk stays SigLIP-class until Empiricist E8 proves otherwise on your robot stills.

One sentence for leadership: schedule I-JEPA for WM literacy; do not swap $\pi_{0.5}$ vision towers without a probe card.

Lineage payoff — where the brick goes

CLEAR subsequent only

**I-JEPA
2023**

Image JEPA
multi-block

**V-JEPA
2024**

Video feature
prediction

**V-JEPA 2
2025**

1M+ h video
scale

2-AC

<62 h DROID
Franka CEM

Cite rule

Any robot success-rate / Franka / DROID claim → cite V-JEPA 2-AC (arXiv 2506.09985), never this paper's experiments.

Must numbers — stakeholder card

Say these aloud once

79.3%

H/14 linear

73.3%

H/14 1%

81.1% / 77.3%

H/16@448

54.2

multi-block T6

66.9 >> 40.7

emb >> pixels

72.4

Clevr Dist

~10×

vs MAE compute

<72 h

16×A100

Exam bite — Stakeholder

One paragraph for Nov midterm

I-JEPA is required for the world-model track because it shows you can get semantic + local features by predicting embeddings (not pixels) with multi-block masks — no view augs — at $\sim 10\times$ lower GPU-h than MAE H/14 (79.3% linear, <72 h on $16\times A100$). VLAs still ship SigLIP/DINO; JEPA's robotics payoff is V-JEPA 2-AC planning, not a drop-in PaliGemma tower. Decision: required WM literacy; optional VLA bake-off.

Numbers tattoo — pin these

Shared across all five seats

79.3%

ViT-H/14 linear

73.3%

H/14 · 1% labels

81.1% / 77.3%

H/16@448 lin / 1%

54.2

multi-block (T6)

66.9 >> 40.7

emb >> pixels (T7)

72.4

Clevr Dist (T4)

~10×

cheaper than MAE H/14

<72 h

16×A100 H/14

<1200

GPU-hours H/14

87.1 vs 87.8

FT H/16@448 vs MAE

no view augs

single-view JEPA

2-AC not here

robots → cite 2506.09985

Stakeholder — close

Takeaways + exam bite

1. Buy: foundational brick → V-JEPA 2-AC; <72h; dual competence; open code; 3-way taxonomy.
2. Risks: image-only; FT -0.7pp; mask hypersensitivity; VLAs chose SigLIP/DINO.
3. Decision: required for world-model track; optional for VLA backbone bake-offs.
4. Cite 2-AC (2506.09985) for robot claims — never this PDF.

EXAM BITE

Required WM literacy; optional VLA tower swap — JEPA robotics = 2-AC planning, not SigLIP replacement.